

NCERT & State Board

Foundation Notes

Class 6-12 Summary Notes + 75 Practice MCQs

Science • History • Geography • Polity • Economy •
Environment

24

Pages

75

Questions

3

Languages

For the foundation stage of MPSC / UPSC / competitive exams

Subject-wise summaries of core concepts from NCERT & Maharashtra State Board textbooks

Every answer program-verified • Answer key at the end

⚠ Honest note: This is a **self-made study summary** based on NCERT / State Board textbooks — not a copy of official books.
For syllabus & exam pattern, always check ncert.nic.in / mpsc.gov.in.

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Physics — Core Concepts (Class 6-12 Summary)

- Speed = distance ÷ time; SI unit m/s
- Acceleration = rate of velocity change; m/s^2
- Newton's 1st law — law of inertia
- 2nd law: $F = m \times a$
- 3rd law: action-reaction equal & opposite
- Gravity $g \approx 9.8 \text{ m/s}^2$
- Moon gravity = 1/6 of Earth
- Work $W = \text{force} \times \text{displacement}$; unit joule
- $\text{KE} = \frac{1}{2}mv^2$; $\text{PE} = mgh$
- Power $P = \text{work} \div \text{time}$; unit watt
- Friction opposes motion; lubricants reduce it
- Pressure $P = \text{force} \div \text{area}$; unit pascal
- Archimedes — buoyant force = weight of displaced fluid
- Density = mass ÷ volume; water 1000 kg/m^3
- Light speed $\approx 3 \times 10^8 \text{ m/s}$ in vacuum
- Refraction — light bends changing medium
- Sound $\approx 340 \text{ m/s}$ in air; none in vacuum
- Ohm's law: $V = I \times R$
- 7 SI base units — metre, kg, second, A, K, mol, cd
- $0^\circ\text{C} = 273 \text{ K} = 32^\circ\text{F}$; 3 temperature scales

✂ Rapid Revision — quick scan

- | | |
|---|---|
| ✂ Weight = $m \times g$; mass stays constant | ✂ Uniform velocity — same direction & speed |
| ✂ g is higher at the poles | ✂ Law of conservation of energy |
| ✂ Newton — unit of force | ✂ Pendulum — $\text{PE} \rightleftharpoons \text{KE}$; period $T \propto \sqrt{L}$ |
| ✂ Joule — work; watt — power; pascal — pressure | ✂ Electric power $P = V \times I$; bill in kWh |

Ancient Indian History

- Palaeolithic → Mesolithic → Neolithic ages
- Indus Valley Civilization — Harappa, Mohenjo-daro
- Harappa excavated 1921 — Daya Ram Sahni
- Great Bath — Mohenjo-daro; Lothal — port
- Indus people knew bronze, not iron
- Rigveda — oldest of 4 Vedas
- 16 Mahajanapadas — Magadha strongest
- Buddha — born 563 BCE Lumbini; enlightenment Bodh Gaya
- Buddhism — 4 Noble Truths & 8-fold path
- Mahavira — 24th Tirthankara; Parshvanatha 23rd
- Chandragupta Maurya — 322 BCE; Chanakya (Arthashastra)
- Ashoka — Kalinga war 261 BCE; edicts in Brahmi
- Gupta era — golden age; Samudragupta (Prayag Prashasti)
- Aryabhata — zero/astronomy; Kalidasa — Abhijnanashakuntalam
- Sanchi Stupa; Ajanta-Ellora caves

✂ Rapid Revision — quick scan

- | | |
|--|---|
| ✂ Indus = Bronze-age urban civilization | ✂ Harsha — Kannauj; Kadambari by Banabhatta |
| ✂ Chanakya — Arthashastra (also Vishnugupta) | ✂ Saraswati — holiest river in Rigveda |
| ✂ Kalidasa — Gupta court poet (navaratna) | ✂ Ajanta — Buddhist mural paintings |
| ✂ Buddhist architecture — stupa | ✂ Mehrgarh — early farming settlement |

Practice S1 — Science (15 Questions)

 Practice S1 — Science (15 Questions) — key on page 24

- Which organelle is called the powerhouse of the cell?
A) माइटोकॉन्ड्रिया **B)** केन्द्रक **C)** रायबोसोम **D)** हरितलवक
- Newton's second law states —
A) $P = W/t$ **B)** $F = m \times a$ **C)** $E = mc^2$ **D)** $V = IR$
- What is the chemical formula of water?
A) NaCl **B)** NH₃ **C)** H₂O **D)** CO₂
- Speed of light in vacuum is approximately?
A) 340 m/s **B)** 1500 m/s **C)** 3×10^6 m/s **D)** 3×10^8 m/s
- How many bones are there in an adult human body?
A) 206 **B)** 238 **C)** 186 **D)** 306
- Deficiency of Vitamin C causes which disease?
A) रिकेट्स **B)** स्कर्वी **C)** रातांधळेपणा **D)** बेरी-बेरी
- Malaria spreads through which mosquito?
A) क्युलेक्स **B)** माशी **C)** अँनोफिलीज **D)** एडीज
- Which gas do plants release in photosynthesis?
A) कार्बन डायऑक्साइड **B)** नायट्रोजन **C)** मिथेन **D)** ऑक्सीजन
- Who developed the first (smallpox) vaccine?
A) एडवर्ड जेनर **B)** फ्लेमिंग **C)** पाश्चर **D)** किटासाटो
- Ohm's law is —
A) $W = Pt$ **B)** $V = I \times R$ **C)** $F = m \times a$ **D)** $P = V \times t$
- The ozone layer protects us from —
A) उल्कापिंडांपासून **B)** आम्लवृष्टीपासून **C)** अतिनीळ (UV) किरणे **D)** सौर वाऱ्यांपासून
- Normal human blood pressure is —
A) 80/120 mmHg **B)** 100/60 mmHg **C)** 140/100 mmHg **D)** 120/80 mmHg
- Brass is an alloy of —
A) तांबे व झिंक **B)** तांबे व कथील **C)** लोहे व कथील **D)** अॅल्युमिनियम व झिंक
- The 10% energy law in ecosystems was given by —
A) व्हिटेकर **B)** लिंडेमन **C)** डार्विन **D)** मेंडेल
- How many chambers does the human heart have?
A) 3 **B)** 6 **C)** 4 **D)** 2

□ Answer Key — All Practice Sets (S1-S5)

Practice S1 — Science (15 Questions)

1→A · 2→B · 3→C · 4→D · 5→A · 6→B · 7→C · 8→D · 9→A · 10→B · 11→C · 12→D · 13→A · 14→B · 15→C

□ Q.1→A (ATP ऊर्जा येथे तयार होते) · Q.4→D (ध्वनीवेग 340 m/s असल्याने फसवू नका) · Q.6→B (A→रातांथळेपणा, B1→बेरी-बेरी, D→रिकेट्स) · Q.7→C (एडीज → डेंग्यू/चिकुनगुन्या) · Q.9→A (वर्ष 1796) · Q.13→A (कांसे = Cu + Sn) · Q.15→C (2 अलिंद + 2 निलोट)

Practice S2 — History (15 Questions)

1→D · 2→A · 3→B · 4→C · 5→D · 6→A · 7→B · 8→C · 9→D · 10→A · 11→B · 12→C · 13→D · 14→A · 15→B

□ Q.1→D (लोथल = बंदर) · Q.3→B (जन्म लुंबिनी, पहिले उपदेश सारनाथ) · Q.4→C (युद्धानंतर बौद्ध धर्म स्वीकारला) · Q.5→D (कुतुबुद्दीन ऐबक — गुलाम वंश) · Q.6→A (मुघल साम्राज्याची सुरुवात) · Q.7→B (रायगडावर; जन्म शिवनेरी 1630) · Q.8→C (A.O. ह्यूम; पहिले अध्यक्ष बॅनर्जी) · Q.9→D (जलियांवाला 1919, भारत छोडो 1942) · Q.11→B (आर्य समाज 1875, ब्राह्मो 1828) · Q.13→D (ऑगस्ट क्रांती) · Q.14→A (ऐबकांनी सुरुवात केली)

Practice S3 — Geography (15 Questions)

1→C · 2→D · 3→A · 4→B · 5→C · 6→D · 7→A · 8→B · 9→C · 10→D · 11→A · 12→B · 13→C · 14→D · 15→A

□ Q.1→C (1 रशिया, 2 कॅनडा, 3 चीन, 4 USA) · Q.2→D (~2525 किमी) · Q.4→B (महाराष्ट्र दिन) · Q.5→C (6 महसूली विभाग) · Q.6→D (1646 मी; अहिल्यानगर जिल्हा) · Q.7→A (8598 मी; एव्हरेस्ट नेपाळात) · Q.8→B (उगम त्र्यंबक, नाशिक) · Q.10→D (जगातील लांब बंधांपैकी एक) · Q.11→A (सर्वात खोल खाई — मेरिना) · Q.12→B (आफ्रिका खंडात) · Q.15→A (गुजरात ते मिझोराम)

Practice S4 — Polity & Economy (15 Questions)

1→B · 2→C · 3→D · 4→A · 5→B · 6→C · 7→D · 8→A · 9→B · 10→C · 11→D · 12→A · 13→B · 14→C · 15→D

□ Q.1→B (स्वीकार 26 नोव्हें 1949) · Q.2→C (घटनासमिती अध्यक्ष — राजेंद्र प्रसाद) · Q.3→D (आंबेडकर युगात नव्हते) · Q.4→A (DPSP भाग 4, कर्तव्ये 4A) · Q.5→B (घटनात्मक उपायांचा हक्क) · Q.6→C (कायम सभागृह) · Q.7→D (कलम 110) · Q.8→A (कलम 148) · Q.9→B (74 वी — नगरी संस्था) · Q.10→C (राष्ट्रीयकरण 1949) · Q.11→D (101 वी घटनादुरुस्ती) · Q.12→A (MPC मध्ये 6 सदस्य) · Q.13→B (वार्षिक वित्तीय विवरण) · Q.14→C (इंदिरा गांधी सरकार) · Q.15→D (योजना आयोगाच्या जागी)

Practice S5 — Mixed Mock (15 Questions)

1→A · 2→B · 3→C · 4→D · 5→A · 6→B · 7→C · 8→D · 9→A · 10→B · 11→C · 12→D · 13→A · 14→B · 15→C

□ Q.1→A (वर्ष 1928) · Q.2→B (1936, उत्तराखंड) · Q.3→C (शून्याचे संशोधन) · Q.4→D (1837 मध्ये) · Q.5→A (पित्त स्रवते) · Q.6→B (थायरॉइड) · Q.7→C (वन्यजीव कायदा 1972) · Q.8→D (बुध, शुक्र, पृथ्वी) · Q.10→B (2017 पासून) · Q.12→D (प्रशांत महासागरात) · Q.14→B (डि. उदयकुमार) · Q.15→C (डेव्हिस कप = टेनिस)